

Exercises from Jan 31 & Feb 8

Ex 7 $[E:F][K] = [E:F][F:K] = [E:F][E:K]$

So $[F:K]$ & $[E:K]$ divide $[E:F][K]$ so $[F:K][E:K]$ divides $[E:F][K]$.
But $[E:F][K] \leq [E:F][F:K]$ hence $[E:F][K] = [E:K][F:K]$.

Ex 8 Let $\alpha_1, \dots, \alpha_n$ be the roots of $f(x)$. Clearly $r \leq n$.

Then $K = K(\alpha_1, \dots, \alpha_r)$ and so it suffices to show that $\text{Irr}(\alpha_i, K(\alpha_1, \dots, \alpha_{i-1}), X) = f_i \in K(\alpha_1, \dots, \alpha_{i-1})[X]$ has degree at most $n - (i-1)$

This is clear as f_i divides $\text{Irr}(\alpha_i, K, X) / (X - \alpha_1) \dots (X - \alpha_{i-1})$ which

has degree at most $\deg(f(x)) - \deg((X - \alpha_1) \dots (X - \alpha_{i-1})) \leq n - i + 1$.

Ex 10 $x^4 = 5 \Rightarrow (ix^2)^2 = -x^4 = -5$

a) $x^2 + 5 = (x + ix^2)(x - ix^2)$ so $\mathbb{Q}(ix^2) =$ splitting field of $x^2 + 5$ is normal over $\mathbb{Q}$

b) $(x + ix)^2 = x^2 + 2ix^2 - x^2 = 2ix^2$ so $\text{Irr}(x + ix, \mathbb{Q}(ix^2), X) = x^2 - 2ix^2$

$= (x - (x + ix))(x + (x + ix)) \Rightarrow \mathbb{Q}(x + ix) =$ splitting of $x^2 - 2ix^2$ over $\mathbb{Q}(ix^2)$

c) $x = x + ix \quad x^2 = 2ix^2 \quad x^4 = -4x^4 = -20 \Rightarrow \text{Irr}(x + ix, \mathbb{Q}, X) = x^4 + 20$

(irreducible by Eisenstein's criterion with $p=5$)

$x^4 + 20 = (x + i\sqrt{5}\sqrt{2})(x - i\sqrt{5}\sqrt{2})(x + i\sqrt{5})(x - i\sqrt{5})$ are roots of $x^4 + 20$

are $\sqrt{20} \cdot \rho^{\pm 1}$

$\rho = \sqrt[4]{\frac{20}{2}} = i\sqrt[4]{\frac{20}{2}}$ is primitive 8-th root of 1.

It is easy to see that not all of these roots are in $\mathbb{Q}(x + ix)$ so $\mathbb{Q}(x + ix)$ is not normal over $\mathbb{Q}$.

(Alternatively $x + ix \neq -x + ix$ are distinct roots of $x^4 + 20$

but $\mathbb{Q}(x + ix) \neq \mathbb{Q}(-x + ix)$ so $\exists \sigma: \mathbb{Q}(x + ix) \xrightarrow{\text{aut}} \mathbb{Q}$ with $\sigma \mathbb{Q}(x + ix) = \mathbb{Q}(-x + ix)$)

$\mathbb{Q}(x + i)$ a) $\mathbb{Q}(i\sqrt{2})$ deg 2 b) $\mathbb{Q}$ deg 1 c) $x^3 - 2 = (x - \sqrt[3]{2})(x - \omega\sqrt[3]{2})(x - \omega^2\sqrt[3]{2})$

$\alpha = (-\frac{\sqrt{2}}{2} + i\frac{\sqrt{3}}{2}) \cdot \sqrt[3]{2} \quad [\mathbb{Q}(\alpha, \sqrt[3]{2}) : \mathbb{Q}] \leq 6$ by Ex 8. $[\mathbb{Q}(\alpha, \sqrt[3]{2}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] [\mathbb{Q}(\alpha) : \mathbb{Q}] = 3 \cdot 2 = 6$

but $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$ & $\alpha \notin \mathbb{Q}(\sqrt[3]{2})$ so $[\mathbb{Q}(\alpha, \sqrt[3]{2}) : \mathbb{Q}(\sqrt[3]{2})] > 2$

so $[\mathbb{Q}(\alpha, \sqrt[3]{2}) : \mathbb{Q}] = 6$

d) $[\mathbb{Q}(\alpha, \sqrt[3]{2}) : \mathbb{Q}] = [\mathbb{Q}(\alpha, \sqrt[3]{2}) : \mathbb{Q}(\alpha, \sqrt[3]{2})] [\mathbb{Q}(\alpha, \sqrt[3]{2}) : \mathbb{Q}] \stackrel{\text{Ex 8}}{=} 6 [\mathbb{Q}(\alpha, \sqrt[3]{2}) : \mathbb{Q}] = 12$

but $\sqrt[3]{2} \notin \mathbb{Q}(\sqrt[3]{2})$ easy check so $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$

α not real so $\alpha \notin \mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{2})$ so $[\mathbb{Q}(\alpha, \sqrt[3]{2}, \sqrt[3]{2}) : \mathbb{Q}] = 12$

$$\Sigma x^{11} \quad e) \quad x^2 + x + 1 = (x - (\frac{1+i\sqrt{3}}{2})) (x - (\frac{1-i\sqrt{3}}{2}))$$

$$[Q(i\sqrt{3}):Q] = 2$$

f) $x^6 + x^3 + 1$ by Σx^5 the roots are $\alpha, \alpha^2, \alpha^4, \alpha^5, \alpha^7, \alpha^8$
 where $\alpha^3 = 1$ eg $\alpha = \cos 40^\circ + i \sin 40^\circ$.

So the splitting field is just $Q(\alpha)$ and $[Q(\alpha):Q] = 6$

g) $x^5 - 7$ has roots $\sqrt[5]{7} \cdot \alpha^i$ $\alpha = \cos \frac{360}{5} + i \sin \frac{360}{5}$ a primitive 5th root of 1 and $i = 0, 1, 2, 3, 4$.

The splitting field is $Q(\sqrt[5]{7}, \alpha)$ (as $\alpha = \frac{\sqrt[5]{7} \cdot \alpha}{\sqrt[5]{7}}$)

Now $[Q(\sqrt[5]{7}):Q] = 5$ as $x^5 - 7$ is irreducible by Eisenstein.

$[Q(\alpha):Q] = 4$ as $f_{\alpha}(x, Q, x) = x^4 + x^3 + x^2 + x + 1$ (check that this is irreducible)

By $\Sigma x^7 \quad [Q(\sqrt[3]{7}, \alpha):Q] = 20$.

$\Sigma x^{12} \quad f: K \rightarrow K$ the Frobenius homomorphism ($\text{Recall } f(x) = x^p$)

$\ker f = 0$ so f is onto and injective. So $f(x) = x$ has a unique solution for all $x \in K$.

Σx^{14} Pick $\alpha \in L$, then $[L:K] = [L:K(\alpha)][K(\alpha):K]$ so $[K(\alpha):K]$ is prime to p . By Prop 6.1 $[K(\alpha):K] = [K(\alpha):K]_s$.

Σx^9 ~~x is a root of $x^9 - 1$ then $\alpha^9 = 1 \Rightarrow \alpha^9 = 1 \Rightarrow \alpha^9 = 1$~~
 $x^{p^3} - 1 = (x^9 - 1)^{p^3} \Rightarrow$ splitting field is $\mathbb{Z}/p\mathbb{Z}$

$\Sigma x^{19} \quad F \subset R \subset E$ Pick $\alpha \in R$ then α is alg. over F so $F[\alpha] \subset R$ is a field so $\alpha^{-1} \in F[\alpha] \subset R$ so R is a field.

But $Q \subset R = Q(x) \subset Q(x)$ & $Q[x]$ is not a field.

Σx^{20} a) let $y = f(x)/g(x) \in K \setminus F$ then $g(x)y - f(x) = 0$ so $F[x]$ is a nontrivial

polynomial eqn in $K[t]$ s.t. $g(x)y - f(x) = 0$ so $F[x]$ is algebraic over K so $K[x] = K(x) = F(x)$.

b) deg $g(t)y - f(t) = n$. It suffices to show $g(t)y - f(t)$ is irreducible in $F(t)[t]$. But y is transcendental, so $F(t)[t] = F[y, t]$ and this is

Ex 20 continued:

In order to check that $g(t)y - f(t)$ is irred. in $F(y)[t]$ it suffices to check that it is irreducible in $F[y][t]$ (see IV 2.2). But deg $_y (g(t)y - f(t)) = 1$ so this can only factor as $p(t)(g_1(t)y - f_1(t))$ but then $p(t) \mid \text{GCD}(g(t), f(t)) = 1$ so $p(t)$ is a unit $\square$

Ex 16. Show $K = F$ α is sep $\iff K(\alpha) = K(\alpha^{p^m}) \forall m > 0$

Proof Clearly $K(\alpha) \supset K(\alpha^{p^m})$. Suppose $\alpha \notin K(\alpha^{p^m})$ then $(X^{p^m} - \alpha^{p^m}) = (X - \alpha)^{p^m}$ so $\text{Irr}(\alpha, K(\alpha^{p^m}), X)$ divides $(X - \alpha)^{p^m}$ hence has 1 distinct root. So $K(\alpha)$ is not sep over $K(\alpha^{p^m})$ hence $K(\alpha)$ is not separable over K .

Converse By the proof of V.6.1, we have $f(x) = h(X^{p^m})$ α^{p^m} is a root of $h(x)$ and $h(x)$ is separable over K . So $K(\alpha^{p^m})$ is sep over K . But $K(\alpha) = K(\alpha^{p^m})$ and so $K(\alpha)$ is separable over K .